MUSEUM ARTIFACT EXPLORER — FINAL REPORT

Flask + PostgreSQL on Supabase (psycopg2) Cloud DBMS Hardening

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1. EXECUTIVE SUMMARY
The Museum Artifact Explorer was fully migrated from local SQLite to cloud-hosted PostgreSQL on Supabase using plain SQL via psycopg2-binary. Database integrity is enforced with GENERATED ALWAYS AS IDENTITY, TIMESTAMP/DATE data types, CHECK, NOT NULL, and UNIQUE constraints, Many-to-Many associations (curators, exhibitions, exhibition_artifacts), a PL/pgSQL AFTER DELETE audit trigger, multi-column indexes, and a denormalized SQL view (artifact_details). Row Level Security (RLS) is enabled across all tables to protect against unauthorized Supabase PostgREST exposure. The test suite runs with complete schema-level isolation.

2. DELIVERABLES MATRIX
• Phase 1 (Database Connection & Wrapper):
  - database.py connects via DATABASE_URL using python-dotenv (sslmode=require).
  - Built DatabaseWrapper and CursorWrapper with psycopg2.extras.DictCursor.
  - Transparently translates '?' parameter placeholders to '%s'.
  - Dict rows support both row['col'] and row[0] indexing; keep commit/rollback/close.

• Phase 2 (PostgreSQL Schema Conversion):
  - Replaced AUTOINCREMENT with INT GENERATED ALWAYS AS IDENTITY PRIMARY KEY.
  - Set created_at to TIMESTAMP DEFAULT CURRENT_TIMESTAMP; exhibition dates to DATE.
  - Created PL/pgSQL function log_deleted_artifact() and trg_artifact_deleted trigger.
  - Replaced SQLite seed logic with INSERT ... ON CONFLICT DO NOTHING.

POSTGRESQL SCHEMA, TEST ARCHITECTURE & SECURITY

3. FINAL RELATIONAL SCHEMA OVERVIEW
• civilizations (id INT IDENTITY PK, name TEXT UNIQUE, region, period, description)
• categories (id INT IDENTITY PK, name TEXT UNIQUE, description)
• artifacts (id INT IDENTITY PK, name NOT NULL, description NOT NULL, period,
  discovery_location, discovery_year INT, material, image_url, civilization_id INT FK,
  category_id INT FK, created_at TIMESTAMP, CHECK(discovery_year <= 2026),
  CHECK(length(trim(name)) > 0), UNIQUE(name, civilization_id))
• favorites (id INT IDENTITY PK, artifact_id INT FK UNIQUE ON DELETE CASCADE, created_at)
• curators (id INT IDENTITY PK, name NOT NULL, email NOT NULL UNIQUE, specialization)
• exhibitions (id INT IDENTITY PK, title NOT NULL UNIQUE, start_date DATE, end_date DATE,
  curator_id INT FK ON DELETE SET NULL, CHECK(end_date >= start_date))
• exhibition_artifacts (exhibition_id INT FK, artifact_id INT FK, PRIMARY KEY, CASCADE)
• deleted_artifacts_log (id INT IDENTITY PK, artifact_id INT NOT NULL, name, deleted_at)
• View: artifact_details (CREATE OR REPLACE VIEW joining artifacts, civs, categories)
• Trigger: trg_artifact_deleted (PL/pgSQL AFTER DELETE trigger inserting to audit log)
• RLS: ALTER TABLE <table> ENABLE ROW LEVEL SECURITY on all 8 tables

4. TEST ARCHITECTURE & ISOLATION
• Dynamic Test Schema: setUpClass creates a dedicated temporary schema (test_museum_<uuid>).
• Search Path Switching: Dynamically executes SET search_path TO "<test_schema>", public.
• Zero Real Data Touch: Tests run exclusively within the temporary test schema.
• Automated Teardown: tearDownClass executes DROP SCHEMA IF EXISTS ... CASCADE.